N+ Network Security

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# Assignment-5

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1. **Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.**




2. **Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).**

### Comparison of Three WAN Technologies

| WAN Technology | Speed | Cost | Real-Life Usage |
|---|---|---|---|
| MPLS | High | High | Connecting Zomato offices in different cities. |
| Leased Line | High and Dedicated | Very High | Connecting large company offices with a dedicated network. |
| Broadband Internet | Medium to High | Low | Used in homes for Wi-Fi and internet access. |

**Conclusion:** MPLS connects multiple offices, Leased Line provides a dedicated connection, and Broadband provides normal internet access at a low cost.

3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.Hint: Focus on how the VPN creates a private tunnel over the public WAN.

**A VPN creates a private encrypted tunnel over a public WAN (Internet).**

1. Connect your phone to the coffee shop's public Wi-Fi.
2. The VPN creates an encrypted tunnel between your phone and the VPN server.
3. Your data travels safely through this tunnel.
4. The VPN server connects you to apps like YouTube or Spotify.
5. Hackers cannot easily read your encrypted data, even if they capture it.

**Conclusion**: A VPN helps protect your data and provides a more secure internet connection when using public Wi-Fi.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

**Answer:**

MPLS (Multiprotocol Label Switching) would be the best WAN technology for Flipkart to connect its warehouses in Ahmedabad, Mumbai, and Delhi.

**Reasons:**

1. **Secure Connection: MPLS provides a private network to connect warehouses securely.**
2. **Reliable Communication: It provides fast and reliable communication between different warehouse locations.**